DR. RVR NRI INSTITUTE OF TECHNOLOGY

An Institution deemed to be University u/s 3 of UGC Act, 1956
School of Computer Studies • Department of Computer Science & Engineering
Pothavarappadu (V), Agiripalli (M), Vijayawada Rural, Eluru District, AP - 521212

ACTIVITY-BASED LEARNING (ABL) ASSIGNMENT REPORT

Course: Web Development Lab (I B.Tech) | **Academic Session:** 2026-2027

Circular Ref: Web Development ABL / Dussehra Holidays Assignment (15-10-2026)

Student Full Name:	Yaswanth	**Academic Year / Sem:**	I B.Tech (2026-27)
Roll Number:	**26000918**	**Branch / Discipline:**	CSE (Computer Science)
Class Section:	**CSM1**	**Repository Name:**	WD-Activity-Based-Learning

Prescribed Case Studies Selection Overview

No.	Case Study Title	Core Web Technologies	Folder Location
1	Student Course Registration Portal	HTML5, CSS3 Grid, DOM State, Credit Meter, Modal Receipt	Case_01_Student_Course_Registration/
2	Campus Event & Tech-Fest Hub (TechVyuha)	Event Filtering, Dynamic Forms, Fee & Pass Generation	Case_02_Campus_Event_Management/
3	Online Book Store & Digital Central Library	Live Search, Cart/Bag Management, 14-Day Due Dates	Case_03_Online_Bookstore_and_Library/

Assignment Objective & Scope:

As communicated in the institutional circular dated 15-10-2026, students are instructed to design and build fully working web applications covering client-side architecture. Each solution incorporates semantic HTML5, responsive CSS3 layouts, interactive JavaScript DOM manipulations, form validation, and downloadable/printable confirmation receipts tailored for authentic university academic operations.

Case Study 1: Student Course Registration Portal

Directory: Case_01_Student_Course_Registration/ | **Primary Files:** index.html, style.css, app.js

1. Problem Statement & Academic Context:

In university credit-based academic systems (such as CBCS at Dr. RVR NRIIT), undergraduate students in the School of Computer Studies must enroll in courses for the upcoming semester while strictly adhering to institutional workload limitations (Minimum 16 credits, Maximum 24 credits). The goal of this case study is to provide an interactive, intuitive course enrollment portal that calculates real-time credit points, dynamically assesses tuition/lab dues, verifies prerequisites, and provides a formal semester registration slip.

2. Key Functional & Technical Implementations:

• **Live Credit Counter & Progress Bar:** Dynamic progress calculation with automated alerts preventing enrollment exceeding 24 credits.
• **Category Filtering:** Instant client-side filtering between Core Subjects, Professional Electives, and Laboratory Courses.
• **Fee Breakdown Calculation:** Real-time computation of tuition fees and laboratory consumables based on selected courses.
• **Official Registration Slip Modal:** Generation of a formal digital slip with Roll No: **26000918**, Student Name: **Yaswanth**, and Section: **CSM1** with print capability.

3. Visual Execution Screenshots:

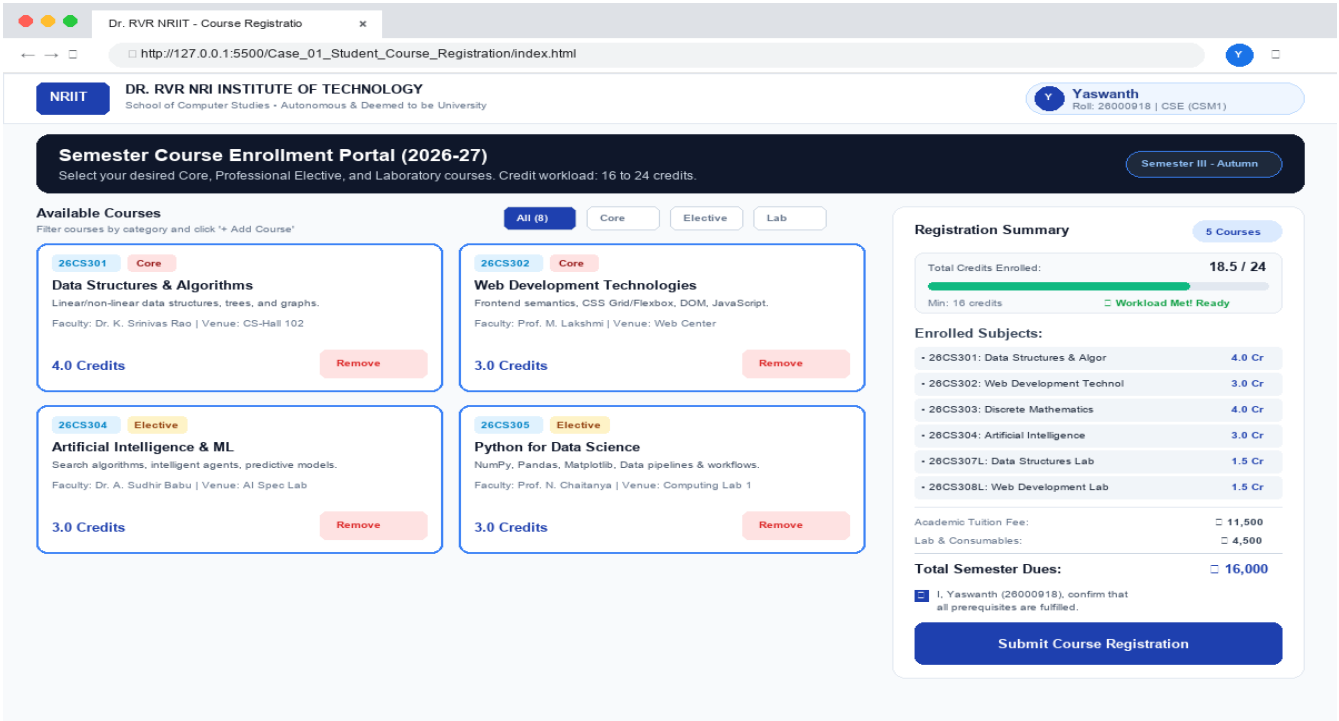


Figure 1.1: Case Study 1 — Course Catalog, Category Filter & Live Credit Workload Meter

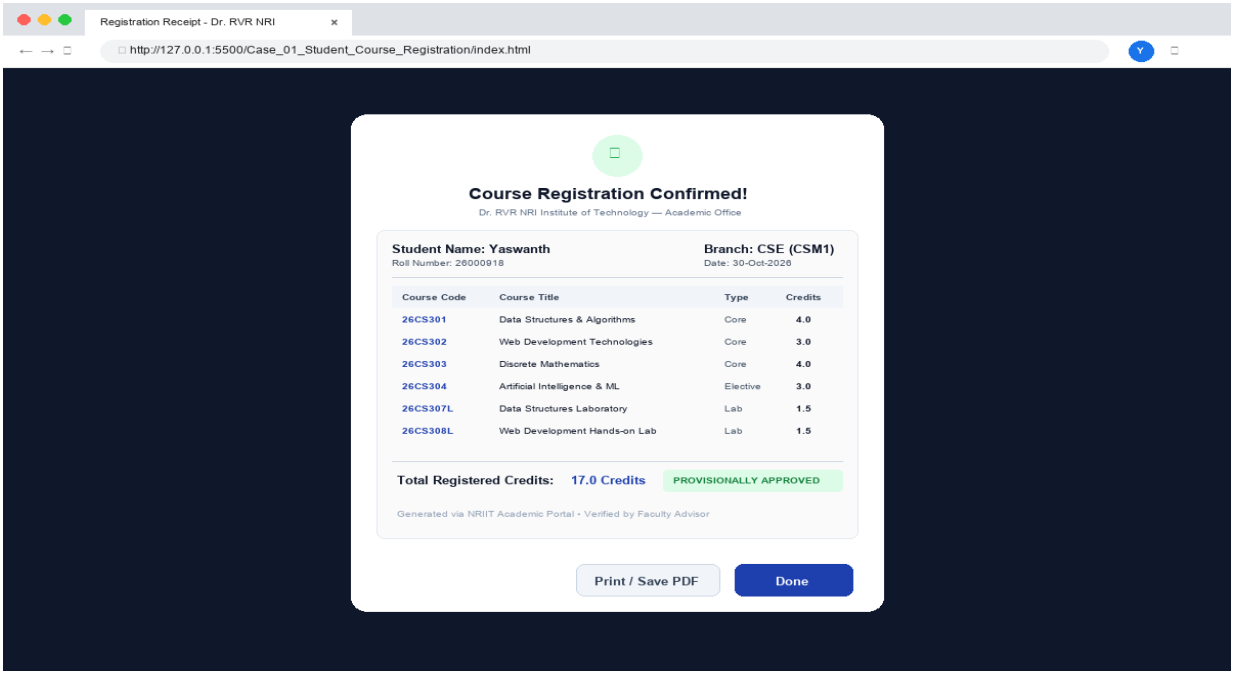


Figure 1.2: Case Study 1 — Verified Registration Confirmation Slip Modal Dialog

Case Study 2: Campus Event & Tech-Fest Hub (TechVyuha '26)

Directory: Case_02_Campus_Event_Management/ | **Primary Files:** index.html, style.css, app.js

1. Problem Statement & Functional Goals:

Annual technical symposiums like *TechVyuha '26* require a high-throughput event discovery and registration platform where students can explore technical hackathons, speed coding contests, paper presentations, and gaming arenas. The platform must handle dynamic team formation, track seat availability in real time, collect participant credentials, and issue verified digital event passes with unique pass IDs.

2. Key Technical Implementations:

• **Live Multi-Attribute Filtering & Search:** Instant real-time search by keywords, venues, and category chips.
• **Adaptive Team Member Generation:** Dynamically generates team member input rows according to each event's maximum team size.
• **Seat Inventory Management:** Updates available slot tallies on successful registration.
• **Digital E-Ticket Pass:** Renders an official tech-fest pass with QR-code placeholder, unique pass ID (e.g. NRIIT-TV26-8918), venue timings, and candidate verification badge.

3. Visual Execution Screenshots:

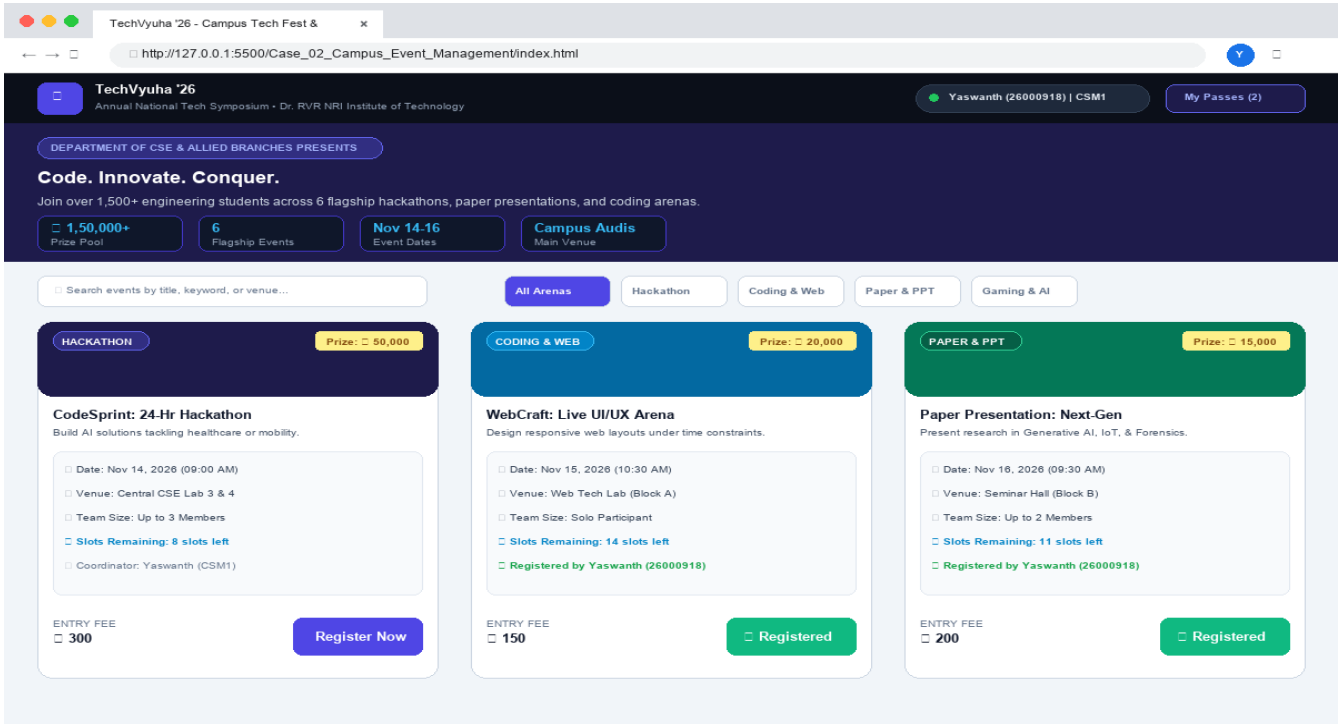


Figure 2.1: Case Study 2 — TechVyuha '26 Event Hub with Live Filters and Prize Badges

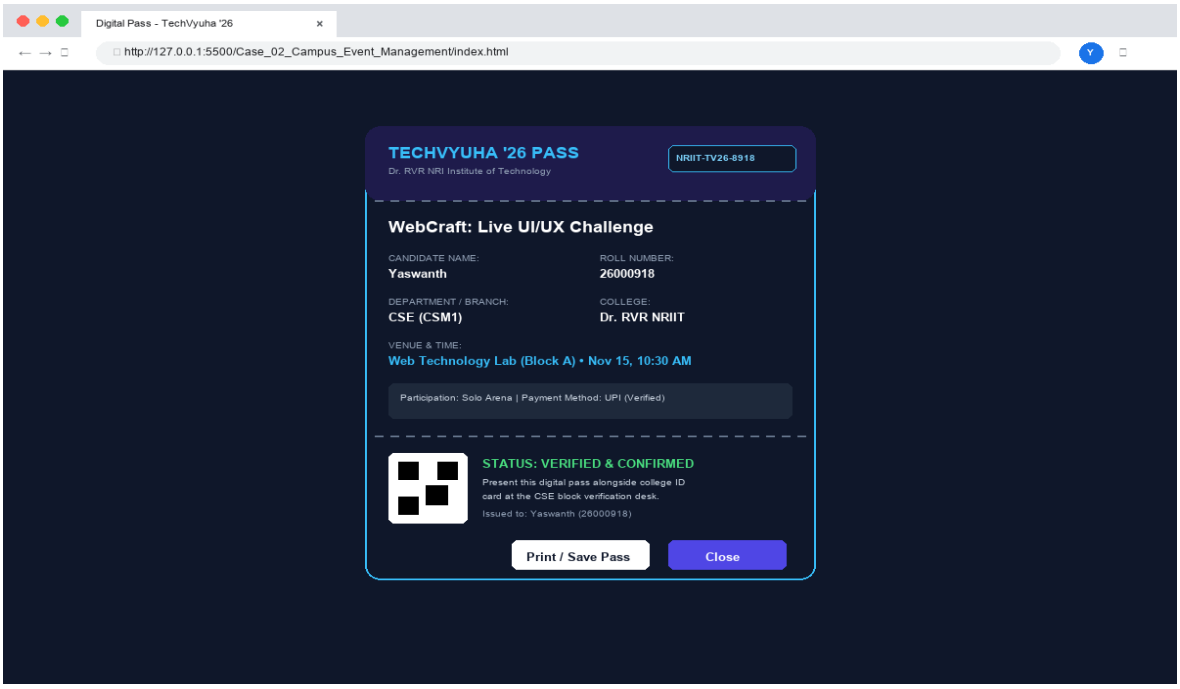


Figure 2.2: Case Study 2 — Generated Verified Digital Entry Pass with Unique Pass ID & QR

Case Study 3: Online Book Store & Digital Library Catalog

Directory: Case_03_Online_Bookstore_and_Library/ | **Primary Files:** index.html, style.css, app.js

1. Problem Statement & Library Workflow:

Academic institutions maintain extensive inventories of technical reference books across diverse engineering disciplines. The Dr. RVR NRIIT Central E-Library application enables students to browse textbooks, check physical shelf rack locations, manage a borrowing cart (Reading Bag) constrained to 3 books, enforce 14-day return due dates, and track active loans.

2. Key Technical Implementations:

- **Real-time Book Search & Discipline Chips:** Search catalog by book title, author, or ISBN.
- **Reading Bag Drawer:** Sliding side drawer managing borrowed volumes with capacity validation (Max 3 books).
- **Automated Due Date Calculation:** Computes exact 14-day return deadlines and outlines late fee regulations.
- **Circulation Receipt & Active Loan Tracker:** Generates library borrowing receipts and allows one-click book returns.

3. Visual Execution Screenshots:

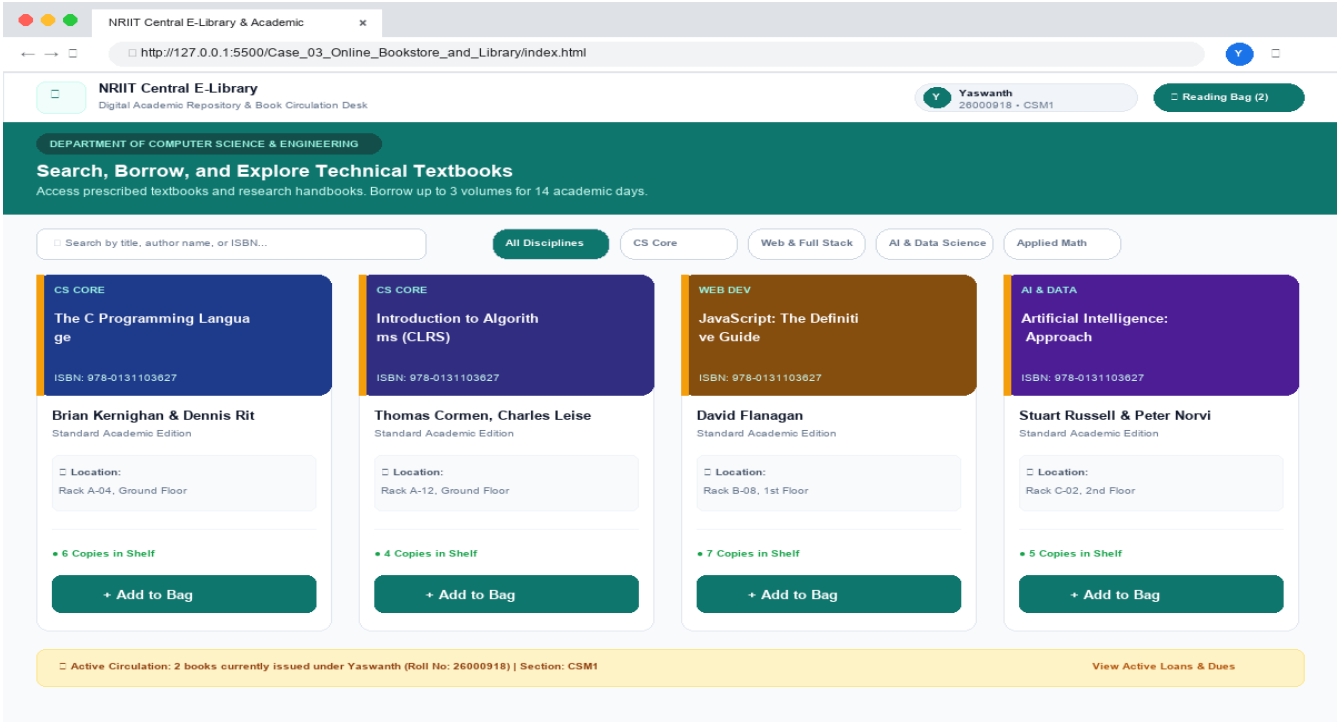


Figure 3.1: Case Study 3 — Central E-Library Catalog with Shelf Locations & Discipline Chips

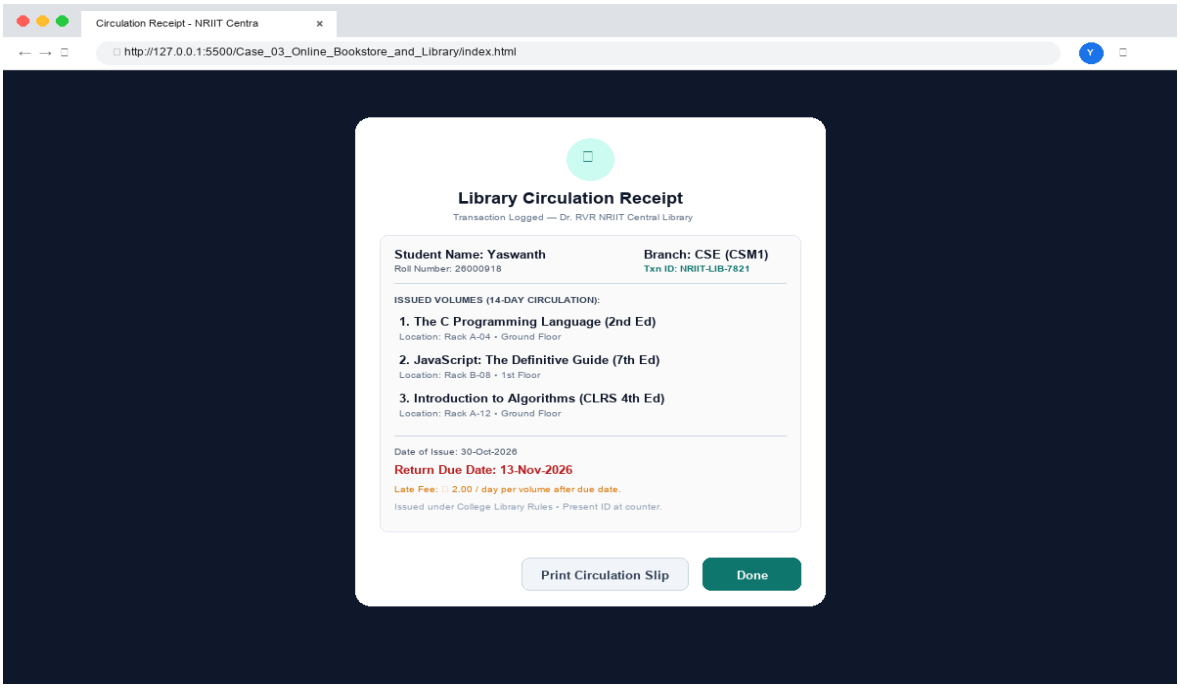


Figure 3.2: Case Study 3 — Official 14-Day Library Circulation Receipt Dialog

Repository Organisation & GitHub Submission Guide

GitHub Repository Name: WD-Activity-Based-Learning

In strict compliance with **Instruction #3** of the circular, all code, screenshots, and this consolidated report are organized inside a clean GitHub repository. The folder structure is laid out as follows:

```
WD-Activity-Based-Learning/
■■■ Case_01_Student_Course_Registration/
■   ■■■ index.html        (Semantic layout, filters, course grid, summary & modal slip)
■   ■■■ style.css         (Custom styling, responsive CSS Grid/Flexbox, cards & badges)
■   ■■■ app.js            (Dynamic rendering, credit limit validation, fee calculation)
■■■ Case_02_Campus_Event_Management/
■   ■■■ index.html        (Hero banner, event arenas, search bar, registration modal)
■   ■■■ style.css         (Tech-fest aesthetic, event cards, digital pass styling)
■   ■■■ app.js            (Filtering, dynamic team fields, unique Pass ID generation)
■■■ Case_03_Online_Bookstore_and_Library/
■   ■■■ index.html        (Textbook catalog, reading bag drawer, circulation receipt)
■   ■■■ style.css         (Academic design, book spine covers, drawer transitions)
■   ■■■ app.js            (Catalog search, 3-book limit, 14-day due date computation)
■■■ screenshots/          (High-resolution browser execution screenshots)
■■■ reports/
■   ■■■ WD_ABL_Consolidated_Report.pdf  (Official consolidated submission PDF)
■■■ README.md             (Student details, selected case study titles & setup guide)
```

Student Declaration & Verification:

I hereby declare that the three case studies presented in this report and uploaded to the GitHub repository WD-Activity-Based-Learning were designed and developed independently by me as part of the Web Development Activity-Based Learning (ABL) course assignment during the Dussehra vacation. All code, design structures, and logic are authentic and ready for live demonstration before the faculty evaluation committee.

Student Signature:	**Faculty / Mentor Evaluation:**
(Yaswanth) Roll No: **26000918** Section: **CSM1** | Branch: **CSE** Dr. RVR NRI Institute of Technology	Marks Awarded: ____ **/ 10 Marks** Signature: _____ Date: _____